

⑤ Select data points to send to next stump.

Salary	Credit	Approval	Bin Assignment
$\leq 50k$	B	No	0 - 0.08
$\leq 50k$	G	Yes	0.08 - 0.16
$\leq 50k$	G	Yes	0.16 - 0.24
$> 50k$	B	No	0.24 - 0.32
$> 50k$	G	Yes	0.32 - 0.40
$> 50k$	N	Yes	0.40 - 0.90
$\leq 80k$	N	No	0.90 - 0.98

• Iterate process selecting random value b/w 0 and 1

Salary	Credit	Approval	Random
$\leq 50k$	N	Yes	0.50
$\leq 50k$	G	Yes	0.10
$\leq 50k$	N	Yes	0.60
$> 50k$	N	Yes	0.75
$\leq 50k$	G	Yes	0.24
$> 50k$	B	No	0.32
$> 80k$	N	Yes	0.87

⑥ These records will be sent to next DT stump.

Salary	Credit	Approval	Sample weight
$> 50k$	N	Yes	1/6
$\leq 50k$	G	Yes	1/6
$> 50k$	N	Yes	1/6
$> 50k$	N	Yes	1/6
$\leq 50k$	G	Yes	1/6
$> 50k$	B	No	1/6
$> 80k$	N	Yes	1/6

T.E (Total Error)

Performance Stump = 0.65

$$F_1 = \alpha_1 (M_1) + \alpha_2 (M_2)$$

$$\alpha_2 = 0.65$$

$$\alpha_1 = 0.896$$

Note: if we calculate you will get α_2
 $\approx \frac{1}{2} \ln(5) \approx 0.805$